

DL-3002: Data Mining -Lab

Lab Task 1

Saad Munir



**Department of Computer Science,
National University of Computer and Emerging Sciences,
Islamabad**

Lab Task: Analyzing Movie and FIFA Dataset

Dataset: 1000 Movies Data.csv & Fifa.csv

Task 1: Data Loading and Exploration

- Load the datasets into a Pandas DataFrame.
- Display the first 10 and last 10 rows of each dataset.
- Explore the basic statistics of the dataset, including mean, median, standard deviation, etc.
- Check for missing values and handle them appropriately (Look out for the role of null values, does removing them will help you or not).

Task 2: Data Manipulation

- Filter out top 10 movies on the basis of their 'Rating' and 'Votes' (Movies Dataset).
- Identify the top 10 movies with the highest profit (Movies Dataset).
- Is there any relation between the movie's rating and revenue? Do movies with higher ratings always have higher revenues (Movies Dataset).
- Find the average runtime of movies directed by a specific director (Movies Dataset).
- Identify the genre with the highest average rating (Movies Dataset).
- Calculate the total revenue generated by movies released in a particular year (Movies Dataset).
- Identify the director who has directed the most movies (Movies Dataset).
- Explore the correlation between runtime and rating. Is there any pattern indicating that longer movies receive higher ratings? (Movies Dataset).
- Find the movie with the longest runtime and its details (title, director, year, etc.) (Movies Dataset).
- Give out the highest paid footballers from 'Germany' (Fifa Dataset).
- What is the average height of French Footballers? (Fifa Dataset).

Task 3: Data Visualization

- Create a histogram to visualize the distribution of movie durations.
- Explore the distribution of movie ratings. Create a histogram to visualize the frequency of different rating ranges.
- Plot a bar chart showing the average rating for the top 10 genres in the dataset.
- Plot a bar chart showing the world highly paid footballers.
- Create a bar chart showing world most fastest footballers (top 10).

Submission:

1. **Naming Convention: “ID_F.Name_L.Name_Section_Task.No”**
2. Submit the Jupyter Notebook or Python script containing your code and visualizations.
3. Provide explanations for each step and your observations.
4. Include any insights or patterns you discover during the analysis.

This lab task covers a range of data analysis and visualization techniques, challenging students to apply their knowledge of Pandas, NumPy, and Matplotlib to a real-world dataset.